BUSINESS PROPOSAL

UMHackathon 2026

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1. Executive Summary:

Homie is an AI-driven rental platform designed to simplify the residential leasing experience in Malaysia. By using GLM-5.1 as a central reasoning engine, the system automates the discovery process by scraping and aggregating data from multiple platforms like ibilik, iProperty, PropertyGuru, and Facebook. The purpose of Homie is to eliminate the stress of manual searching by transforming messy, fragmented listings into a single, organized stream of verified results.

The project targets a major market opportunity in the Klang Valley, where high demand from young professionals and students creates a Total Addressable Market (TAM) of RM 2.8 million. Homie's core competitive advantage is its autonomous agentic workflow. Unlike traditional search engines, our dual-level ReAct orchestration allows the AI to handle the entire process, from understanding a user's specific needs through interactive chat to initiating direct, automated outreach to landlords on Telegram.

The business model generates revenue through success-based commission on successful rentals, B2B subscription plans for property agents, and the sale of market intelligence reports derived from aggregated rental data to property agents and landlords.

2. Product Overview / Solution

Key Features:

- **Conversational Requirement Intake:** An AI chat agent that uses natural language to understand and extract your specific rental needs like budget, location, and pet preferences.

- **Unified Multi-Source Discovery:** Automatically scrapes and aggregates live listings from ibilik, iProperty, PropertyGuru, and Facebook simultaneously.

- **Explainable Match Scoring:** A deterministic engine that ranks every house from 0 to 100 based on eight dimensions, including price fit and transport proximity.

- **Autonomous Telegram Outreach:** A direct AI agent that initiates negotiations with landlords via Telegram to verify unit availability and requirements.

- **Live Workflow Transparency:** A real-time progress feed that shows the user exactly what the AI is "thinking" and doing at every stage.

- **Public Transport Verification:** Uses Google Maps to show the exact location of LRT stations and bus stops so you can see if the transport links are actually nearby.

- **Neighborhood Feedback:** Imports Google reviews for the building or area so you can check the "real vibes" and feedback from current residents.

Technology Used:

- AI Orchestration Layer: Powered by ILMU-GLM-5.1 using a dual-level ReAct loop to manage complex task planning and decision-making.

- High-Performance Backend: Built with Python 3.11 and FastAPI using asynchronous workers to handle multiple scraping and AI tasks at once.

- Modern Frontend: A Next.js 14 dashboard using Tailwind CSS v4 and shadcn/ui for a clean, professional user experience.

- Robust Data Ingestion: Uses Playwright for browser automation on JS-heavy sites and httpx + BeautifulSoup for high-speed static scraping.

- Direct Communication Stack: Integrated with Telethon (MTProto) to allow the AI agent to communicate directly through the Telegram API.

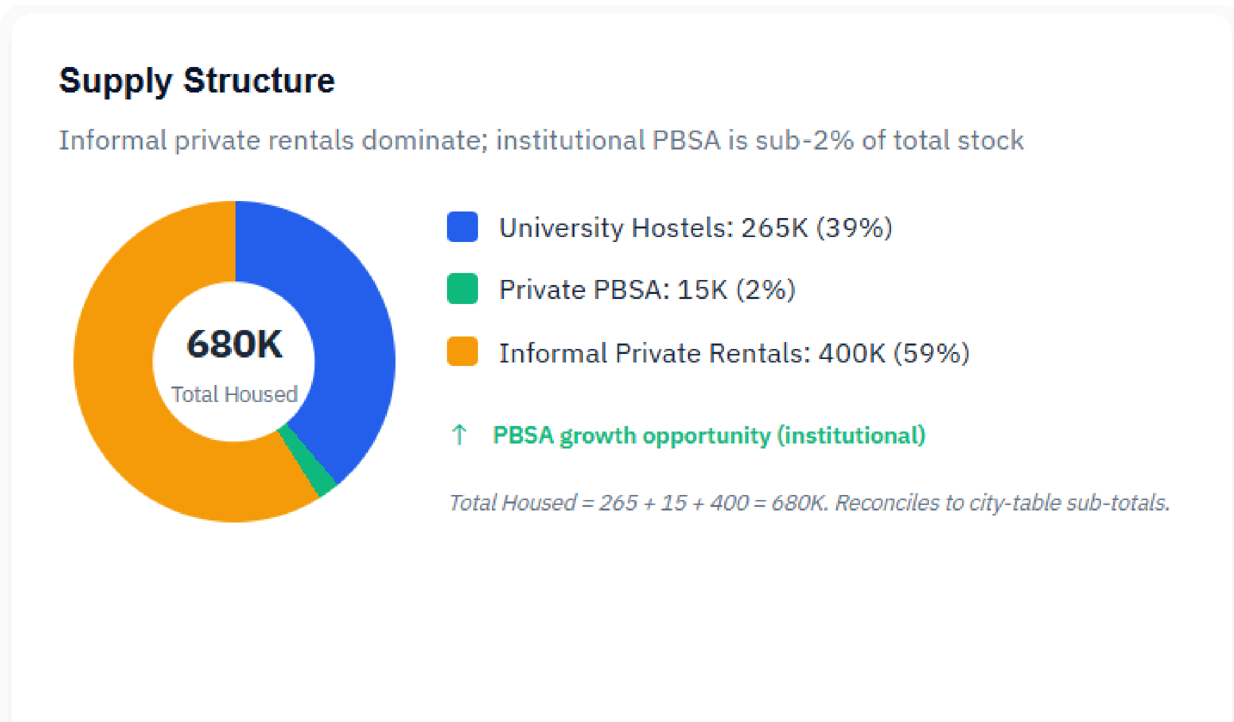
- Scalable Data Store: Utilizes PostgreSQL for primary data persistence and SQLAlchemy for efficient database management.

3. Market Research & Market Dynamics

3.1 Problem Landscape

*Renters in Malaysia are forced to **manually search across multiple disconnected platforms** each with inconsistent listing formats, varying price conventions, and no unified way to compare options. This fragmented landscape **wastes hours of a tenant's time**, **makes fair price assessment nearly impossible**, and often results in missed opportunities due to **information overload**. From the perspective of rentals, they **lack a unified digital platform** to search and reach out to houses that fit their requirements. The limitation rises when they are new in the rental market and have to find a comfortable place to settle down within a short time. Besides, for landlord agents, they **lack digital infrastructure** to reach wider customer bases and smoothen the operations instead of repetitively posting in various platforms and tracking user rental interest.*

3.2 Market Niche



***Pie chart 1** shows 59% of university students opt for Informal Private Rentals.*

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Supply by City

Two distinct metrics shown side-by-side: **Formal Provision %** (the institutional benchmark, formal beds only) and **Total Housed %** (all students with shelter, including informal private rentals). Institutional PBSA investment decisions reference the former.

City	Students	University Hostels	Private PBSA	Informal Rentals	Formal Provision % (Hostel + PBSA) ÷ Students	Total Housed % All shelter, incl. informal
Kuala Lumpur	350,000	70,000	8,000	150,000	22%	65%
Selangor	280,000	80,000	4,000	100,000	30%	66%
Penang	120,000	35,000	1,500	40,000	30%	64%
Johor Bahru	150,000	40,000	1,000	50,000	27%	61%
Other States	200,000	40,000	500	60,000	20%	50%
TOTAL	**1,100,000**	**265,000**	**15,000**	**400,000**	**24%**	**62%**

***Table 1** shows the number of students opting for informal rentals across Malaysia exceeds university hostels.*

***Context 1:** For most Malaysians under 35 renting a home in Klang Valley, the numbers currently favour renting — not buying. That is not a cultural opinion. It is arithmetic. At 2026 property prices and interest rates, the monthly cost of owning the same unit you can rent is typically 20–40% higher, before factoring in the RM100,000–RM120,000 you need upfront just to get the keys. (Alicia. 2026, April 23. Rent or Buy Property in Malaysia? The 2026 Decision Guide. SPEEDHOME. https://speedhome.com/blog/rent-vs-buy-malaysia/)*

***Context 2:** As of early 2026, the active rental vacancy rate in Malaysia's major urban condo markets is estimated at 8% to 15%, with prime Kuala Lumpur pockets closer to 6% to 10% and oversupplied suburban high-rises at 12% to 18%.*

Across different neighborhoods in Malaysia, vacancy rates range from under 8% in high-demand areas like Bangsar and Mont Kiara to over 15% in older investor condo stock in less connected suburban locations. (William Tan. 2025, December 31. Malaysia Property Industry Review 2025: Key Trends, Challenges, and What to Expect in 2026. PEPS. https://www.peps.org.my/malaysia-property-industry-review-2025-key-trends-challenges-and-what-to-expect-in-2026/#:~:text=According%20to%20national%20market%20data,continued%20to%20experience%20overhang%20pressures%20.)

***Context 3:** As of early 2026, well-priced mid-market condos in Malaysia's Klang Valley typically stay listed for about 20 to 45 days before finding a tenant.*

Across different property types and neighborhoods in Malaysia, days on market range from as few as 10 to 25 days for student-oriented rooms near universities, to 45 to 75 days or longer for high-end rentals above RM7,500 per month. (William Tan. 2025, December 31. Malaysia Property Industry Review 2025: Key Trends, Challenges, and What to Expect in 2026. PEPS. https://www.peps.org.my/malaysia-property-industry-review-2025-key-trends-challenges-and-what-to-expect-in-2026/#:~:text=According%20to%20national%20market%20data,continued%20to%20experience%20overhang%20pressures%20.)

Markets for demand:

1. *Referring to **Pie chart 1** and **Table 1** above extracted from Malaysia Student Housing Report 2026, the surge in university student enrolment has led to a shortage of on-campus housing, forcing many off-campus students to seek alternative accommodations nearby. Hence, Homie targets university students across Malaysia especially in Kuala Lumpur and Selangor area which have more than 40 higher education institutions. **The target segment is those students who prioritize matching requirements, convenience, speed and digital experience.***
2. *Based on **Context 1**, most Malaysians prefer renting a house than buying a house. **The market segment is for people working in Malaysia especially the Kuala Lumpur and Selangor area who prefer long-term renting options that suit their requirements, speed and price.***

Markets for supply:

1. *Based on **Context 2** and **3**, the **national structural vacancy rate** (including units that are built but never occupied) sits closer to **20%** and for high-end units (renting for above RM7,500), landlords often face "empty house" periods of **45 to 75 days or longer**. This shows that the difficulty in finding tenants is one of the major issues faced by landlord agents. Hence, **the market segment is landlord agents or companies who are also looking to expand their reach without their own digital infrastructure.***

3.3 Market Sizing

Market Tier	Definition	Estimated Value
TAM	**Total Reachable Digital Rental Commissions.** 1% of the rental expenditure for the urban "Under-35" demographic nationwide.	**RM 2.8 Million** 10% commission on 40k tenant matches from 100k active user

SAM	**Mid-Term (3 Years)** All university towns and industrial hubs across Malaysia (Perak, Melaka, Penang, etc.).	**RM 1.4 Million** 10% commission on 20k tenant matches from 60k active user
SOM	**Short-Term (1–1.5 Years)** KL & Selangor area for students and the general working public.	**RM 175k** 10% commission on 2500 tenant matches from 30k active user

3.4 Market Dynamics & Placement:

There were few key dynamic that shape Homie's entry to this market:

- **Smartphone Penetration:** *The smartphone penetration in Malaysia is accelerating, with the market projected to grow from 10704.25 USD Billion in 2025 to 16734.95 USD Billion by 2035, exhibiting a CAGR of 4.5% during the forecast period 2025 - 2035. So, eventually it accelerated the consumer comfort primarily connected with application-based service.*
- **AI-recommentation and Summary Momentum:** *Recent data from the 2026 Digital Economy Trust Index and local research indicates that while 78% still use traditional search engines like Google for verification, 44% now prefer AI-embedded search for initial discovery because it reduces the manual effort of clicking through multiple links.*
- **Trust AI Through Transparency:** *86% of Malaysian consumers state that they trust AI recommendations more when the brand is* ***transparent*** *about using AI. They do not necessarily want a human but they want a reliable assistant that saves them time.*
- **Digitalization of Landlord Agents Platform:** *The agents were also looking for digital exposure and distribution channels over the traditional manually posting in various platforms and consistently posting to avoid potential market and vendors overlook in this information burst era.*

4. Value Proposition & Competitive Advantage

Homie is enhancing prospective tenants' experience when finding a home. It removes tenants' need to browse for endless hours. By scanning and evaluating properties for them, Homie reduces the workload to descriptive prompting and decision making. As a one of a kind agent, Homie is a pioneer in the business . Instead of competing with existing platforms, Homie turns them into a tool to find a property that best suits tenants' criterias.

5. Competitor Analysis (Porter's Five Forces) (SWOT Analysis)

Aspect	Key Point	Description
Strengths	Natural Language Processing	Homie accepts human-written content unlike current platforms that filter their search via forms. Users can describe property characteristics more clearly
	Objective Evaluation	Homie's scoring algorithm shows users how each property fits into their preferences.
Weaknesses	Limited Database	Since Homie is a web-scraper, it's output is only as good as it's sources. More platforms will need to be added as Homie's searching grounds. Additionally, property websites do not include certain tenant preferences, hence reducing a property's final score.
Opportunities	Next Generation Adults	Prospective users are most likely Gen Z who are accustomed to the chatbot interface. This makes Homie feel user-friendly.
Threats	Repetitive Listing	Homie's output may contain duplicates as the same property might be listed on multiple platforms.

Competitor Comparison Table:

Competitor	***Strength***	***Weakness***	***Homie's Advantage***
EdgeProp	a. AI mode filters can be used to enhance search efficiency. b. "Send Enquiry" button to contact the agent for a particular property.	Their AI mode provides basic sentiment understanding and does not do well with higher level language.	Telegram bot integration allows auto-messaging with agents.
Rentola	a. Simple but soothing design. b. Displays price of current unit compared to other units. Tells the user whether the property is below market price or otherwise.	Similar UI to existing platforms.	List out potential houses based on the score so manual comparison is not required.

6. Business Model Canvas

6.1. Customer Segmentation

1. **Urban Consumer**: Urban Malaysian tenants with limited time available for property research actively search for rental accommodation (rooms or full units) in Kuala Lumpur and the Klang Valley, particularly those with specific requirements around transport, budget, furnishing, and property policies.
2. **Relocators** : Fresh graduates relocating for employment, expatriates unfamiliar with local rental platforms & second-year university students transitioning from campus hostels to private off-campus housing.
3. **Property agents & landlords** : Property agents or landlords who wish to benchmark their listings' market competitiveness against aggregated, scored results.

6.2. Revenue Stream

Primary : Commission

- We charge a 10% Success Fee of the first month's rent upon a successful tenancy closing initiated through the Homie platform.

Secondary : Credit-Based Plan

- Instead of a rigid monthly subscription, users operate on a freemium model. They receive a basic allowance of daily AI searches for free, but can purchase "Homie Credits" (e.g., RM5 for 50 credits) to unlock premium, on-demand features during their house hunt.

Tertiary : **Market Analysis**

- Landlords and agents use credits to pay for data-driven insights, including competitive benchmarking, demand heatmaps, and pricing optimization reports to improve listing performance.

6.3. Cost Structure

Cost Drivers	Cost Type
• Homie's maintenances of the platform, cloud infrastructure, research & development, product development associated costs	Fixed Cost
• Real-time computational expenses for autonomous data scraping and LLM token consumption.	Variable Cost
• Targeted digital campaigns to reach users, vendor onboarding programs to expand property databases, and the administrative costs of providing user assistance.	Overhead Cost

6.4. Key Partners

Internal	External
Homie Engineering Team	AI & Cloud Infrastructure Providers
Homie AI Development Team	Data Source Platforms (iBilik, iProperty..)

6.5 Risk Assessment

Risk Type	Impact	Likelihood	Mitigation
Scraper Reliability & UI Changes	High	Medium	Implement schema validation at the scraper output boundary. If a source layout changes and fails, the system degrades to show partial results from remaining sources.
AI Hallucinations & Score Inconsistency	High	Low	Use code-based rules rather than relying on the LLM. Enforce null-allowed fields during data normalization so AI doesn't fabricate missing info.
Platform Anti-Scraping / IP Blocking	Medium	High	Operate scrapers at human-browsing request rates.
Fraudulent Listings & Scams	High	Medium	Flag listings where the price is suspiciously below the market average or uses duplicate photos.

7. Implementation & Financials

7.1 Go-to-Market Strategy

Homie will adopt a **Hyper-Local, Phased Rollout** strategy to solve the classic "chicken-and-egg" problem of a two-sided marketplace. Instead of burning capital on a massive nationwide launch, we will build targeted density through three distinct phases:

7.2 Implementation Plan

Phase	Action	Timeline	Target
Student Wave	Hyper-local launch at university zones (e.g. UM, Sunway)	Month 1-3 (3-4 months before new semester)	5,000 student users
Vendor Seeding	B2B onboarding via “Zero-Risk Trial” (free initial leads)	Months 4-8	300 Vendor Partners

National Scaling	Broaden to all major urban centers (Penang, Johor) via digital marketing & referrals.	Month 9-18	30,000 Active Users
TOTAL	Nationwide Penetration	18 Months	Reach Breakeven

7.3 Revenue Projections (e.g 3-Year)

Metric	*Year 1*	*Year 2*	*Year 3*
Active Cities	*3*	*6*	*12*
Successful Tenancy Matches	*1,500*	*8,000*	*20,000*
Partner Vendors	*500*	*1,500*	*4,000*
Commission Revenue	*RM 105,000*	*RM 560,000*	*RM 1.5M*
Credit Revenue	*RM 25,000*	*RM 80,000*	*RM 300,000*
Subscription Analysis Revenue	*RM 12,500*	*RM 37,500*	*RM 100,000*
Total Revenue	*RM 142,500*	*RM 677,500*	*RM 1.9 M*
Net Position	*RM (8,000)*	*RM +180,000*	*RM +0.9M*

7.4 Key Financial Milestones

- Month 4: Complete the initial "Student Wave" pilot in the Klang Valley, reaching 5,000 active users and successfully onboarding the first batch of "Zero-Risk Trial" vendor partners.
- Month 12: Successfully transition pilot vendors to the paid commission/subscription model and expand operations to Penang and Johor, hitting the Year 1 revenue target.
- Month 18: Reach operational break-even. At this point, the monthly recurring revenue from agent subscriptions and match fees covers all API usage and cloud infrastructure costs.
- Month 30: Achieve strong positive cash flow. Reach 50,000 active users nationwide across 12 cities, pacing towards the Year 3 revenue target.